

GENERAL SPECIFICATION G8.1

Building Works

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G8.1 BUILDING WORKS

G8.1.1 Masonry

- (a) Masonry shall be constructed in brickwork, concrete blockwork or stonework as detailed in the Drawings.
- (b) References in this Specification to bricks and bricklaying shall also be applicable to blocks and blocklaying and to stones and stonework as appropriate.
- (c) Brickwork and blockwork which is to be left exposed un-rendered or unplastered either externally or internally is defined as fair faced work.

G8.1.2 Bricks and Blocks

- (a) Bricks, whether facing or common shall be from an approved manufacturer subject to the approval of the S.O.. Building bricks shall comply with the requirements of ASTM C62 - 17 and shall be hard, well-burnt, machine made wire-cut bricks or pressed.
- (b) Facing bricks shall comply with the requirements of ASTM C216-17a. Facing bricks shall be of the colour and texture selected by the S.O..
- (c) All bricks shall be of standard dimensions 215 x 102.5 x 65 mm. Bricks shall be regular in size and shape, with good faces, clean arises, sound and free from cracks and any other defects and contaminants.
- (d) All blocks shall be hard concrete, even in shape, square with true arises, well matured, of a dense well-consolidated mixture and of standard dimensions 390 mm x 190 mm x 100 mm or 190 mm x 90 mm x 100 mm.
- (e) All concrete blocks shall be from an approved manufacturer and shall be in accordance with BS EN 772.
- (f) Load-bearing blocks shall have a crushing stress of not less than 5.6 MN/m² and non-load-bearing blocks shall have a crushing strength of not less than 3.6 MN/m².
- (g) Three quarter, half and quarter size and special corner blocks shall be used for bonding where applicable.

G8.1.3 Samples and Handling

- (a) Samples of each type of brick and block to be used in the Works shall be submitted to the S.O. for his approval before materials are ordered. The approved samples shall be kept on the Site for reference.
- (b) All bricks or concrete blocks shall be carefully stacked and handled upon delivery to Site Any rejected brick/blocks shall be removed from the Site immediately and at the Contractor's own expenses.

G8.1.4 Mortar

- (a) Mortar for masonry shall comply with the requirements of BS EN 998-2 and Eurocode 6: Part 2.

- (b) Mortar shall be machine mixed on site only as required immediately prior to use, and any mix which is over 3/4 hour old shall be rejected. No mortar, which has commenced to set shall be used or mixed with any other mortar.
- (c) Mortar for masonry work shall not be mixed by hand.
- (d) Materials for mortar for brickwork shall be measured by volume and shall consist of 3 parts sand and 1 part cement. The cement shall be stored at the Site in a perfectly dry structure and all consignments shall be used in order of delivery. Cement affected by dampness shall not be used in the Works.
- (e) Plasticising additives shall not be used without prior approval from the S.O..

G8.1.5 Bricklaying

- (a) Unless otherwise specified, brick and blockwork walls shall be constructed in accordance with the recommendations of PD 6697:2010, BS EN 1996-3:2.
- (b) Prior to the laying of any facing brickwork, sample panels 900 mm by 600 mm of each facing brick type shall be built using mortars made with different fine aggregates, white or coloured cements or colouring agents as directed by the S.O.. When the required ingredients for the mortar have been determined and approved by the S.O. they shall be used for all mortar for facing brick and the appropriate test panel shall be retained for reference and shall represent the standard to which all facing brickwork shall conform. Failure to meet the standard of workmanship reflected by the sample panel may result in the affected brickwork panel being condemned by the S.O. and, in this eventuality, the condemned panel shall be demolished and reconstructed at the Contractor's own expense.
- (c) Bricklaying shall be carried out such that cutting of bricks is reduced to the minimum by careful setting out. No bats or broken bricks shall be incorporated into the Works unless essential for bond. Where cut bricks or blocks are required, all cutting shall be carried out using mechanical cutting disks.
- (d) Brickwork shall be set out accurately to the various dimensions, thickness and heights shown on the Drawings. Before commencing work, storey rods showing the heights of all courses shall be set out and these shall be strictly followed.
- (e) All brickwork shall be plumb within 5 mm of plane and level over 3 m in any direction. All work found not to comply strictly with this requirement shall be taken down and rebuilt at the Contractor's own expense.
- (f) Every third course shall be levelled. The setting out and level of individual courses shall be controlled using levelled string guides, two being required (one for each skin) in the case of one brick thick walls constructed in Stretcher bond, and for cavity walls, where walls are specified as fair-faced both sides.
- (g) All bricks shall be well soaked before use and kept damp as necessary and the top of walls left off shall be damp before work re-commences. During hot or dry weather, all bricks shall be immersed in water for 5 minutes immediately prior to laying. All constructed walling exposed to the sun shall be kept damp immediately after its completion and during the following day.

- (h) All walls shall be kept to a uniform height and at no point shall brickwork in the same wall unit be carried up more than five (5) courses above any adjoining portion. No brickwork shall be carried up more than 1.8 m in one day.
- (i) Before commencing any bricklaying, the Contractor shall ensure that all pipes, conduits, bolts, frames or any other materials necessary to be installed in the work have been fixed or provided for.
- (j) Beds and vertical joints of brickwork shall be filled solid with mortar, and no vertical joint shall be allowed to be flushed up from the top but each joint must be filled up as the bricks are laid. All work found not to comply strictly with this requirement shall be taken down and rebuilt at the Contractor's own expense.
- (k) Brickwork shall be finished up neat against the soffits of beams and slabs with bricks cut to correct lengths and laid vertically. Bricks laid on the slant shall not be used.

G8.1.6 Brickwork Bond

- (a) All 102.5 mm ("half brick thick") thick walls shall be built entirely in stretcher bond. All 205 mm ("one brick thick") thick walls shall be built in Flemish bond unless otherwise specified or directed by the S.O..
- (b) All piers and intersections and angles of walls shall be properly bonded together. All meeting brickwalls/blockwork shall be toothed and bonded together at every alternate course.
- (c) All perpend, reveals, quoins and other angles of the walls etc. shall be built strictly true and square. No broken bricks or bats shall be used except where closers are required at quoins, jambs etc to make bond.

G8.1.7 Joints

- (a) Joints shall be broken accurately and the thickness of bed joints shall be no greater than 11.5 mm and not less than 8.5 mm. In addition, brickwork shall be gauged to rise 300 mm in four courses.
- (b) Facing brickwork shall be finish-pointed with weathered joints 3 mm deep formed before the mortar has set, unless the S.O. directs otherwise.
- (c) Brickwork or blockwork for internal surfaces which are not to be rendered or plastered shall be fair faced with a flush joint made as work proceeds.
- (d) Joints in brickwork or blockwork which is to be rendered or plastered shall first be filled solid with mortar as specified and shall then be raked out to a depth of 10 mm whilst the mortar is green to form an adequate key for plaster.
- (e) Separation joints shall be formed by inserting approved joint filler 10 mm thick. 60 days after the bricks have been laid the filler shall be raked out to a depth of 10 mm and the joints shall be pointed with approved elastoplastic sealant.

G8.1.8 Ties

- (a) Wall ties, bonding ties and hold fasts embedded at least partially in the mortar joints of brickwork or blockwork shall be made either of low carbon steel and galvanised in accordance with Clause 5 of BS 1243 or of stainless steel, Grade 304.

- (b) Wall ties shall be provided to effectively bond the two brickwork skins together at the following locations:
 - (i) where facing brick is required for the external face and common brick is required for the internal face of brick walls; and
 - (ii) Elsewhere wherever a 205 mm ("one brick thick") wall is detailed on the Drawings to be constructed in Stretcher bond.
- (c) The wall ties shall be spaced 900 mm apart horizontally, 450 mm vertically and staggered and with extra ties at reveals and openings. The wall ties shall be flats with fishtails at either end and to the approval of the S.O..
- (d) The Contractor shall submit samples of his proposed wall tie and bonding tie system(s) to the S.O. together with the relevant manufacturer's certificates prior to first use and upon approval shall not vary the system(s) without the prior approval of the S.O..

G8.1.9 Bonding between Brickwork and Structures

- (a) The Contractor shall provide bonding ties at spacings of not more than 450 mm vertically, at locations where brickwork butts against concrete columns, steel columns and concrete walls. The bonding ties shall be 3 mm thick galvanised mild steel or stainless steel fishtailed strips 30 mm wide and 180 mm long, unless otherwise directed by the S.O..
- (b) Every bonding tie shall be securely anchored to the structure as follows:

(i) Concrete columns, concrete walls	The bonding tie shall be effectively anchored by slotting the tie fishtail into a galvanised mild steel or stainless steel preformed channel cast into the face of the concrete column or wall and the other half of the tie built into the brickwork. The channel shall be anchored into the concrete with welded lug anchors or studs 60 mm long at 250 mm centres
(ii) Steel columns	The bonding tie shall be effectively anchored by shot firing the tie fishtail to the steel flange or web
(iii) Concrete encased steel columns	The bonding tie shall be effectively anchored to a galvanised mild steel or stainless steel preformed channel cast into the face of the encasement. The channel shall be anchored into the concrete with lug anchors or studs 60 mm long at 250 mm centres

- (c) The channels and bonding ties shall be a proprietary system to the approval of the S.O..

- (d) The Contractor shall submit a layout for the installation of the bonding ties around the perimeter of an individual brickwork panel to the S.O. and shall mark out the tie location for the inspection of the S.O. prior to commencing brick laying for the particular panel.

G8.1.10 Wall Reinforcement

- (a) All brick walls shall be reinforced with a layer of approved brickwork reinforcement set at every fourth course. The width of the reinforcement strip shall be 62 mm for a "half brick" thick brick wall and 165 mm for a "one brick" thick wall, unless otherwise directed by the S.O.. Reinforcement strips for "one brick" thick brick walls constructed with two stretcher bond leaves shall be continuous across the two brickwork leaves.
- (b) All 100 mm thick hollow blockwork walls shall be reinforced with a layer of approved reinforcement at every second course.
- (c) The wall reinforcement shall comprise an expanded galvanised mild steel mesh strip of 0.711 mm thickness (22 SWG), the bottom most reinforcement being 150 mm above the bottom of the individual brickwork panel. The wall reinforcement shall be laid in continuous long lengths and shall have a minimum lap of 350 mm at all reinforcement joints, bent around at angles and intersections and carried through the full thickness of the end wall at corners and junctions. The reinforcement shall be discontinued at movement joints. Where the wall bonds with thicker walls, the reinforcement shall be carried through the thicker wall.
- (d) The wall reinforcement shall be lapped with the concrete column and concrete wall ties.

G8.1.11 Damp-Proof Courses and Flashing

- (a) Three-ply bituminous felt damp proof course complying in all respects with BS 743 Type B or equivalent approved by the S.O. shall be laid for the full thickness in a continuous length one course above ground floor level as shown on the Drawings or as directed by the S.O.. The bituminous felt shall be lapped at least 150 mm at angles and joints laid on a 12 mm thick bed of fresh cement mortar (1:4) the full thickness of the walls. If the mortar has been allowed to go hard the bituminous felt shall be bedded in and coated on the bottom and top surfaces with liquid bitumen.
- (b) All walls shall be carefully cleaned and well wetted before the cement bed for the damp proof course is laid.
- (c) Flashing for brickwork and stone masonry shall comply with the recommendations of PD 6697:2010, BS EN 1996-3:2.

G8.1.12 Lintels and Sills

- (a) Unless otherwise shown on the Drawings all brickwork over openings shall be supported on precast concrete lintels and of the dimension shown on the Drawings or as directed by the S.O..
- (b) Window sills and wall copings of precast concrete, cast stone, clayware and natural stone shall comply with BS 5642.

G8.1.13 Building-In

- (a) Generally, and wherever practical, all items shall be built in as work proceeds. The Contractor shall ensure all built in items are pre-primed as required, set and braced into correct position, have all building-in provisions properly fixed, reinforced to prevent distortion and protected against damage.
- (b) Openings for doors, windows, ventilation grills, etc. shall be properly marked out and left unbuilt until the frames have been fixed in position.
- (c) All frames shall be built in as work proceeds unless otherwise approved by the S.O.. All door frames, window frames and vent frames, plates, etc. shall be bedded in cement mortar (1:4), 10 mm thick.
- (d) All frames shall be carefully fixed and stayed in position and shall be anchored to the walls with 225 mm x 40 mm x 3 mm galvanised steel holdfasts unless otherwise shown on the Drawings, one end turned up 75 mm and twice counter sunk, drilled and screwed to frame, the other end swallow tailed and built into joints of walling. Holdfasts shall be spaced at approximately 600 mm centres unless otherwise shown in the Drawings.

G8.1.14 Protection of Finished Work

- (a) Completed brickwork and blockwork including all decorative elements, finished surfaces, particularly corners and jambs, etc. shall be protected at all times from scaffold splash, mortar droppings, grout leakage from suspended slabs and the harmful effects of the weather as soon as they are built. The protection shall be using timber casing or other suitable means.
- (b) Newly erected brickwork shall be protected against weather to prevent the mortar from being washed out of the joints. Should any brickwork become damaged by sun or rain the brickwork so damaged shall be removed by the Contractor and rebuilt at his own cost.

END OF SECTION